Gioia Zheng

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Research Profile

Applied Computer Science and AI undergraduate at Sapienza University of Rome studying where retrieval-augmented generation fails, how retrieval and generation errors interact, and how evaluation choices affect measured conclusions. Current work centers on information retrieval, RAG evaluation, failure analysis, and reproducible experiments.

Education

Sapienza University of Rome Sep 2022 – Present
B.Sc. in Applied Computer Science and Artificial Intelligence

Research Experience

NLP Research Intern, Microsoft Feb 2026 – May 2026
Natural Language Computing, remote

- Designed and implemented a multi-stage retrieval and generation evaluation pipeline over MS MARCO-scale data: BM25 retrieval, FAISS dense retrieval, transformer reranking, and seq2seq generation.
- Built paired evaluation routines with confidence intervals to measure how stage-level retrieval changes propagate to answer quality.
- Defined schema-versioned experiment manifests and validation checks so results remain traceable to code, data, configuration, and environment.

Selected Research Projects

MS MARCO Generative QA System github.com/GioiaZheng/msmarco-genqa
Python, PyTorch, BM25, FAISS, cross-encoder reranking, T5, paired bootstrap

- Investigate when retrieval and reranking improvements translate into better grounded generation, rather than only higher aggregate surface metrics.
- Built a reproducible retrieve → rerank → generate pipeline on MS MARCO passage data, with config-driven experiments and schema-versioned manifests.
- Reported Token-F1 improvement from 0.197 to 0.368 on 6,980 paired queries, with paired-bootstrap confidence interval [+0.163, +0.178].
- Developing a failure taxonomy that separates retrieval, reranking, evidence-use, and generation errors; dense results are explicitly limited to a qrels-anchored sample.

Handwritten OCR Pipeline github.com/GioiaZheng/handwritten-ocr-system
PyTorch, CNN, BiLSTM, CTC, IAM handwriting dataset

- Implemented a handwritten text recognition pipeline from preprocessing to CRNN/BiLSTM/CTC decoding, with CER/WER evaluation hooks.
- Documented training configuration, deterministic splits, and the open requirement for measured CER/WER reporting rather than presenting unverified results.

Additional Experience

Web Development Intern, Idearia Mar 2026 – Present
Maintain production systems and automate operational workflows in a client-facing environment.

Young Representative, UPI Emilia-Romagna Sep 2022 – Oct 2023
Conducted data-driven research on school dropout and contributed to policy-oriented reports.

Skills

Programming	Python, SQL
ML and Retrieval	Information retrieval, RAG evaluation, PyTorch, FAISS, BM25, Sentence Transformers, cross-encoder reranking, T5
Research Methods	Paired bootstrap, confidence intervals, controlled ablations, grounding evaluation, failure analysis, CER/WER
Research Tooling	Linux, Docker, Git, GitHub Actions, experiment manifests, reproducible pipelines
Languages	Chinese (Native), Italian (Bilingual), English (Professional), German (Beginner)